Exploratory Data Analysis (EDA): Univariate Analysis

Course: 100 Days of Machine Learning (Day 20) **Teacher:** CampusX

1. Introduction to EDA

Exploratory Data Analysis (EDA) is the process of analyzing a dataset to understand its characteristics, often using visual methods. The primary purpose is to "get a feel" for the data and understand it "inside out" before applying any Machine Learning models.

Types of Analysis in EDA:

1. **Univariate Analysis:** Analyzing a single variable (one column) at a time.
2. **Bivariate Analysis:** Analyzing two variables simultaneously to find relationships.
3. **Multivariate Analysis:** Analyzing more than two variables together.

2. Univariate Analysis

The term **Univariate** comes from "Uni" (Single) and "Variate" (Variable).

- **Definition:** When you analyze a single column or variable independently to understand its distribution and nature, it is called Univariate Analysis.
- **Goal:** To gain knowledge about each column in the dataframe individually.

Identifying Data Types

Before starting analysis, you must ask: **"What is the nature of my data?"** Data is generally divided into two types:

1. **Numerical Data:** Numbers (e.g., Height, Weight, Age, Price, Battery Capacity).
2. **Categorical Data:** Groups or Labels (e.g., Nationality, Gender, College Branch, State).

The type of graph or tool you use depends entirely on whether the column is Numerical or Categorical.

3. The Dataset: Titanic

In this video, the teacher uses the **Titanic Dataset** to demonstrate Univariate Analysis.

Key Columns Explained:

- **PassengerId:** Unique ID (Numerical, but not very useful for prediction).
- **Survived:** 0 = Died, 1 = Survived (Categorical).
- **Pclass:** Passenger Class 1, 2, or 3 (Categorical/Ordinal).
- **Name:** Name of passenger.
- **Sex:** Male/Female (Categorical).
- **Age:** Age of passenger (Numerical).
- **SibSp:** Number of siblings/spouses aboard (Categorical/Discrete Numerical).
- **Parch:** Number of parents/children aboard (Categorical/Discrete Numerical).

- **Ticket:** Ticket number.
- **Fare:** Price paid for the ticket (Numerical).
- **Cabin:** Cabin number.
- **Embarked:** Station where passenger boarded (S=Southampton, C=Cherbourg, Q=Queenstown) (Categorical).

4. Univariate Analysis on Categorical Data

A. Count Plot

Used to see the frequency of each category (how many times each value appears).

Code:

```
import seaborn as sns

import matplotlib.pyplot as plt


# Using seaborn to create a count plot

sns.countplot(df['Survived'])
```

Line-by-Line Explanation:

1. import seaborn as sns: Imports the Seaborn library for advanced visualization.
2. sns.countplot(df['Survived']): Creates a bar chart showing the count of people who survived (1) vs. those who did not (0).

Teacher's Observation:

- In the Titanic data, more than 500 people died (0), while less than 400 survived (1).
- You can also use df['Survived'].value_counts() to see these numbers in text format.

B. Pie Chart

Used when you want to see the distribution of categories in terms of **percentage**.

Code:

```
df['Survived'].value_counts().plot(kind='pie', autopct='%.2f')
```

Line-by-Line Explanation:

1. df['Survived'].value_counts(): Counts the occurrences of 0s and 1s.
2. .plot(kind='pie'): Tells pandas to visualize these counts as a pie chart.
3. autopct='%.2f': A formatting string that displays the percentage values on the chart with 2 decimal places.

Teacher's Tip: * Pie charts are great for understanding the "share" of a category (e.g., "61% of passengers died").

- You can use this for Pclass or Sex to see the gender ratio or class distribution.

5. Univariate Analysis on Numerical Data

Numerical data is continuous, so we cannot just count unique values. We look for the **Distribution**.

A. Histogram

A histogram groups numerical data into "bins" (ranges) and shows how many data points fall into each bin.

Code:

```
import matplotlib.pyplot as plt

plt.hist(df['Age'], bins=5)
```

Line-by-Line Explanation:

1. plt.hist(df['Age']): Plots the distribution of the 'Age' column.
2. bins=5: Divides the age range into 5 equal parts. If you increase bins, you get more detail; if you decrease them, the graph becomes "thicker" and less detailed.

Teacher's Reasoning: * Histograms help you see where the majority of the data lies (e.g., most passengers were between 20 and 40 years old).

B. Distplot (KDE Plot)

This is an improved version of a histogram. It shows a curve called **KDE (Kernel Density Estimation)**.

Code:

```
sns.distplot(df['Age'])
```

Line-by-Line Explanation:

1. sns.distplot(df['Age']): Plots the histogram with a smooth line over it.

Key Concept: PDF (Probability Density Function)

- The smooth line is the PDF.
- **The Y-axis represents Probability.**
- **How to read it:** If you pick a random passenger, what is the probability that their age is 40? You look at 40 on the X-axis, go up to the curve, and check the Y-axis value.
- It helps determine if the data is **Skewed** (leaning to one side) or **Normally Distributed** (Bell Curve).

C. Box Plot

Used to find the **5-Number Summary** and identify **Outliers**.

Code:

```
sns.boxplot(df['Age'])
```

The 5-Number Summary Explained:

1. **Minimum:** The lowest value (calculated, not necessarily the smallest in data).
2. **First Quartile (Q1):** 25th percentile (25% of data is below this).
3. **Median (Q2):** 50th percentile (the middle value).
4. **Third Quartile (Q3):** 75th percentile (75% of data is below this).
5. **Maximum:** The highest value (calculated).

Formula for Outliers:

- **IQR (Interquartile Range) = Q3 - Q1**
- Any value **lower than** Q1 - 1.5 * IQR is an outlier.
- Any value **higher than** Q3 + 1.5 * IQR is an outlier.

Teacher's Tip:

- If you see dots outside the "whiskers" of the box plot, those are **Outliers**. For example, in the 'Fare' column, some people paid $500, which is an outlier compared to the average $30 fare.

6. Mathematical Univariate Analysis

Apart from graphs, you can use built-in functions to get quick stats:

- df['Age'].min(): Smallest age.
- df['Age'].max(): Oldest age.
- df['Age'].mean(): Average age.
- df['Age'].skew(): Measures the asymmetry.
  - **0:** Perfectly symmetrical (Normal distribution).
  - **Positive:** Tail is on the right.
  - **Negative:** Tail is on the left.

Summary / Key Highlights

1. **Categorical Analysis Tools:** Countplot and Pie Chart.
2. **Numerical Analysis Tools:** Histogram, Distplot, and Boxplot.
3. **Outliers:** Use Boxplot to identify points that deviate significantly from the rest of the data.
4. **Goal of Univariate Analysis:** Understand the "Spread" and "Center" of each variable independently.

Final Advice: Always start your EDA with Univariate analysis to know your columns before trying to find relationships between them!